

NANGMAGVIK LAKE, NT, Canada

WMO: 719740

Lat: 74.139N

Lon: 119.989W

Elev: 105

StdP: 14.64

Time Zone: -7.00 (NAM)

Period: 96-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-45.4	-42.8	-53.8	0.2	-45.2	-51.0	0.3	-42.5	34.8	-3.1	30.5	-8.4	4.6	150	1.125

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	63.6	55.4	58.3	52.4	53.2	49.0	55.7	62.7	52.5	58.0	49.1	53.0	13.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
51.7	57.1	58.3	48.5	50.7	55.3	45.9	45.8	51.9	23.7	62.6	21.7	58.0	19.8	53.1	63.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.9	26.2	23.1	-47.4	67.8	3.2	4.9	-49.7	71.3	-51.6	74.1	-53.4	76.9	-55.8	80.5		
(5)			-47.5	58.5	3.1	3.5	-49.7	61.0	-51.6	63.1	-53.3	65.0	-55.6	67.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.7	-20.8	-24.3	-20.2	-4.9	17.9	37.2	43.2	39.2	28.3	7.2	-7.5	-17.6		
	DBStd	26.58	11.35	11.80	11.44	10.96	9.74	6.99	7.15	6.63	7.40	11.52	11.21	11.69		
	HDD50	15868	2195	2079	2177	1647	995	392	240	343	651	1327	1725	2097		
	HDD65	21297	2660	2499	2642	2097	1460	835	675	799	1101	1792	2175	2562		
	CDD50	48	0	0	0	0	0	8	30	10	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	10.1	10.4	9.2	9.7	9.3	10.3	11.6	11.7	10.8	11.0	9.4	8.3	9.2		
	PrecAvg	5.9	0.2	0.2	0.3	0.3	0.4	0.5	1.1	1.1	0.7	0.6	0.4	0.2		
	PrecMax	8.1	0.6	0.9	0.6	0.8	1.0	1.2	1.9	2.2	1.4	1.5	0.9	1.0		
(15) Precipitation	PrecMin	3.5	0.0	0.0	0.0	0.0	0.1	0.1	0.4	0.2	0.2	0.1	0.0	0.0		
	PrecStd	1.3	0.1	0.2	0.1	0.2	0.2	0.3	0.4	0.5	0.3	0.3	0.2	0.2		
	DB	12.6	10.8	13.1	30.6	41.8	65.8	69.3	68.6	48.6	33.0	25.9	14.9			
	MCWB	11.9	10.1	12.3	29.0	38.1	55.8	57.5	58.4	45.7	32.7	25.2	14.2			
	2% DB	4.2	2.7	6.2	21.3	37.2	58.2	65.2	59.2	44.0	29.7	17.6	9.4			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	3.5	2.1	5.4	20.3	34.7	51.6	56.2	53.3	42.7	29.2	16.4	8.8			
	5% DB	-0.2	-1.7	1.0	16.6	34.2	51.8	60.8	53.5	40.7	26.9	11.8	4.0			
	MCWB	-0.8	-2.3	0.4	15.6	32.4	46.9	53.3	49.6	39.8	26.1	11.3	3.6			
	10% DB	-4.0	-6.0	-3.7	12.2	31.5	47.4	56.3	49.3	37.5	23.3	7.9	-1.1			
	MCWB	-4.5	-6.5	-4.3	11.4	30.2	43.8	51.2	46.9	36.7	22.6	7.4	-1.7			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	12.2	10.1	12.6	29.0	38.4	56.1	60.0	59.1	45.8	32.9	25.3	14.6			
	MCDB	12.7	10.9	13.2	30.7	41.4	64.4	65.5	67.0	47.5	32.9	26.0	15.0			
	2% WB	3.6	2.4	5.4	20.3	35.0	51.6	56.8	54.1	42.8	29.3	16.9	9.0			
	MCDB	4.2	2.9	6.0	21.1	36.8	57.7	63.9	59.2	44.1	29.6	17.7	9.6			
	5% WB	-0.8	-2.1	0.5	15.8	32.6	47.5	54.1	49.9	39.8	26.2	11.3	3.5			
(33) Mean Daily Temperature Range	MCDB	-0.2	-1.6	1.0	16.6	34.0	51.7	60.7	53.0	40.6	26.8	11.8	4.0			
	10% WB	-4.6	-6.5	-4.2	11.3	30.2	43.9	51.1	46.9	36.8	22.6	7.3	-1.6			
	MCDB	-4.1	-6.0	-3.7	12.1	31.6	47.2	56.1	49.1	37.4	23.2	7.7	-1.1			
	5% DB	12.8	12.8	13.6	15.8	11.6	9.1	11.6	10.0	7.1	11.1	12.8	12.1			
	MCDBR	17.0	15.8	16.9	15.4	12.3	16.0	19.2	17.8	10.1	9.1	14.0	14.2			
(36) Clear-Sky Solar Irradiance	MCWBR	16.9	15.6	16.3	14.6	10.4	10.9	12.0	12.5	8.7	9.2	13.9	14.0			
	5% WB	17.0	16.1	16.6	15.1	11.2	16.1	19.0	16.4	9.8	9.1	14.4	14.1			
	MCWBR	16.8	15.9	16.1	14.4	9.6	11.3	12.7	12.1	8.6	9.1	14.4	14.0			
	taub	N/A	0.164	0.237	0.308	0.326	0.332	0.330	0.321	0.277	0.168	N/A	N/A			
	taud	N/A	1.879	2.133	2.014	2.063	2.288	2.402	2.419	2.345	1.923	N/A	N/A			
(42) All-Sky Solar Radiation	Ebn at Noon	0	111	223	243	258	265	261	246	211	98	0	0			
	Edh at Noon	0	7	19	35	39	32	28	23	16	6	0	0			
(44) All-Sky Solar Radiation	RadAvg	0	42	444	1292	1951	2062	1681	946	395	92	0	0			
	RadStd	0	5	22	63	116	189	124	104	46	14	0	0			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page